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COMP 3008: Human-Computer Interaction
Assignment 4

1.

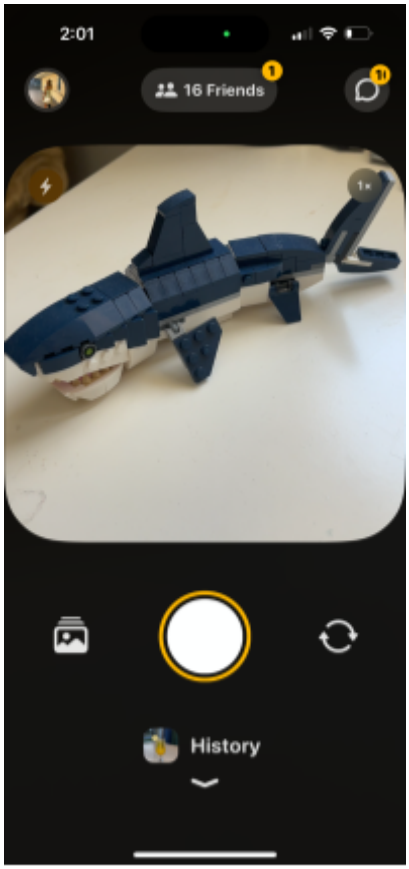


Figure 1: Locket Homescreen

a. Locket Widget App

One task a user might do with this interface is send a photo to a friend. The user opens the Locket widget on their home screen, a unique feature that sets it apart from standard app icons (Figure 2). Then the user taps it to activate the camera, takes a quick selfie or snaps a photo of what they're doing, adds a small emoji or short caption, and hits send. Within seconds, the photo appears directly on their friends' home screens, allowing users to share moments instantly and stay connected.

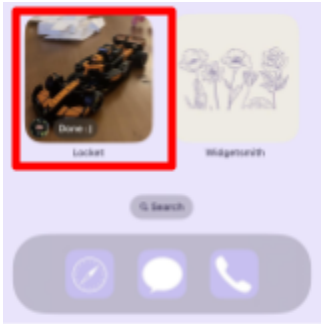


Figure 2: Widget of Locket on home screen of Iphone

b. In the app, an example of an adaptable function is the ability for users to customize their widget layout and choose which friends' photos appear on their home screen. This feature allows users to manually control how their content is displayed and whose updates they prioritize. This adaptability supports usability by enhancing user control and personalization, two key aspects of user-centered design. It gives users the freedom to configure the interface in a way that best fits their preferences and habits, which improves satisfaction and reduces cognitive load since the most relevant content is easily accessible and easy to remember. In this context, adaptability is a good design choice because it allows users to tailor their experience without introducing unnecessary complexity or automation that could interfere with their intent.

An adaptive function in the Locket app could be the system's ability to automatically prioritize recently interacted friends or frequently viewed photos at the top of the feed. This improves usability by minimizing the amount of navigation and search effort required to reach the most relevant content, directly supporting efficiency and cognitive ease. The app automatically responds to user behavior, allowing quicker access to commonly used functions and creating a smoother, more intuitive experience. This adaptivity makes interactions feel more natural and personalized, helping users complete their tasks faster while maintaining focus on the app's main goal, sharing and viewing photos effortlessly.

c. Fifty years into the future, the next generation of the Locket app could be built around the conversing conceptual model, where interaction feels more like a natural dialogue between the user and the system rather than a series of taps and swipes. Instead of manually taking and sending photos, users could verbally instruct the app through an AI conversational assistant integrated into wearable or augmented reality devices. For example, a user might say, "Send this moment to Emma," and the system would capture the user's current visual field through AR glasses, enhance the lighting automatically, and instantly send the image. The system could then respond conversationally with confirmations like, "Photo sent to Emma" This conversational interaction type improves effectiveness by removing unnecessary manual steps and reducing input errors, allowing users to share photos easily in real time while staying present in their environments.

By applying this conversing conceptual model, the interface directly supports the usability goal of effectiveness. Tasks that currently require multiple steps, opening the widget, taking a photo, selecting friends, and sending, can be completed almost instantaneously with natural language commands. Users receive immediate feedback through the system's verbal responses, reinforcing Shneiderman's principle of feedback and ensuring that the system status is always visible. This design also aligns with naturalness and perceived affordances, as speaking a command is intuitive and aligns with users' mental models of everyday communication.

Additionally, this approach could improve learnability for new users. Since the app relies on conversational prompts rather than hidden menus or icons, users can quickly understand how to complete core tasks without extensive trial and error. The verbal dialogue guides users through the process, providing built-in help and reducing cognitive load, which addresses challenges identified in the current interface where users struggle to locate features or understand actions. Overall, using a conversing conceptual model makes the Locket app faster, more intuitive, and more satisfying, while maintaining its core purpose of sharing moments instantly.

d. One major design problem is the lack of clear feedback when users encounter errors, such as receiving a message like "Not enough photos" (User 1 Transcript). The SUS results show an average score of 65, below the benchmark of 68 for a "good" system. Coupled with interview insights. This directly impacts the usability goal of effectiveness, because users are unsure how to resolve the issue, which can prevent them from successfully sending or receiving photos. This violates Shneiderman's principle of feedback and error prevention, as the system does not provide clear, actionable information. This conclusion is supported by observation data, where User 1 hesitated and expressed uncertainty, demonstrating that the interface does not effectively communicate system status.

Another design problem is that essential features, such as the delete option being hidden under "Share" or the auto-send checkmark, are not immediately obvious (User 2 Transcript). This

lowers learnability, especially for new users trying to set up the widget or send a first photo. From a design perspective, this is related to perceived affordances and naturalness, since users cannot easily infer how the interface behaves based on standard conventions or their mental models. Observation of User 2 struggling with these tasks led to the conclusion that the app's structure requires redesign to make features more visible and intuitive.

Based on the observational data, the most significant barrier to the usability goals of effectiveness and learnability is the accidental activation of premium features, which is a classic capture error. This problem arose directly from the Think-Aloud observation, where User 1 unexpectedly navigated to the "Locket Gold" page while trying to perform a core task, expressing confusion with statements like, "Woah, well now I'm at Locket Gold somehow." This violates the consistency and standards design principle where the interface uses familiar icons but assigns them non-standard functions, betraying the user's mental model built from other apps. When a user intends to take a photo but is instead captured by a more frequently practiced habit (tapping a camera-like icon) and routed to a sales page, it directly undermines effectiveness by preventing task completion and severely harms learnability by making the interface feel unpredictable and misleading.

2.

a.

i.

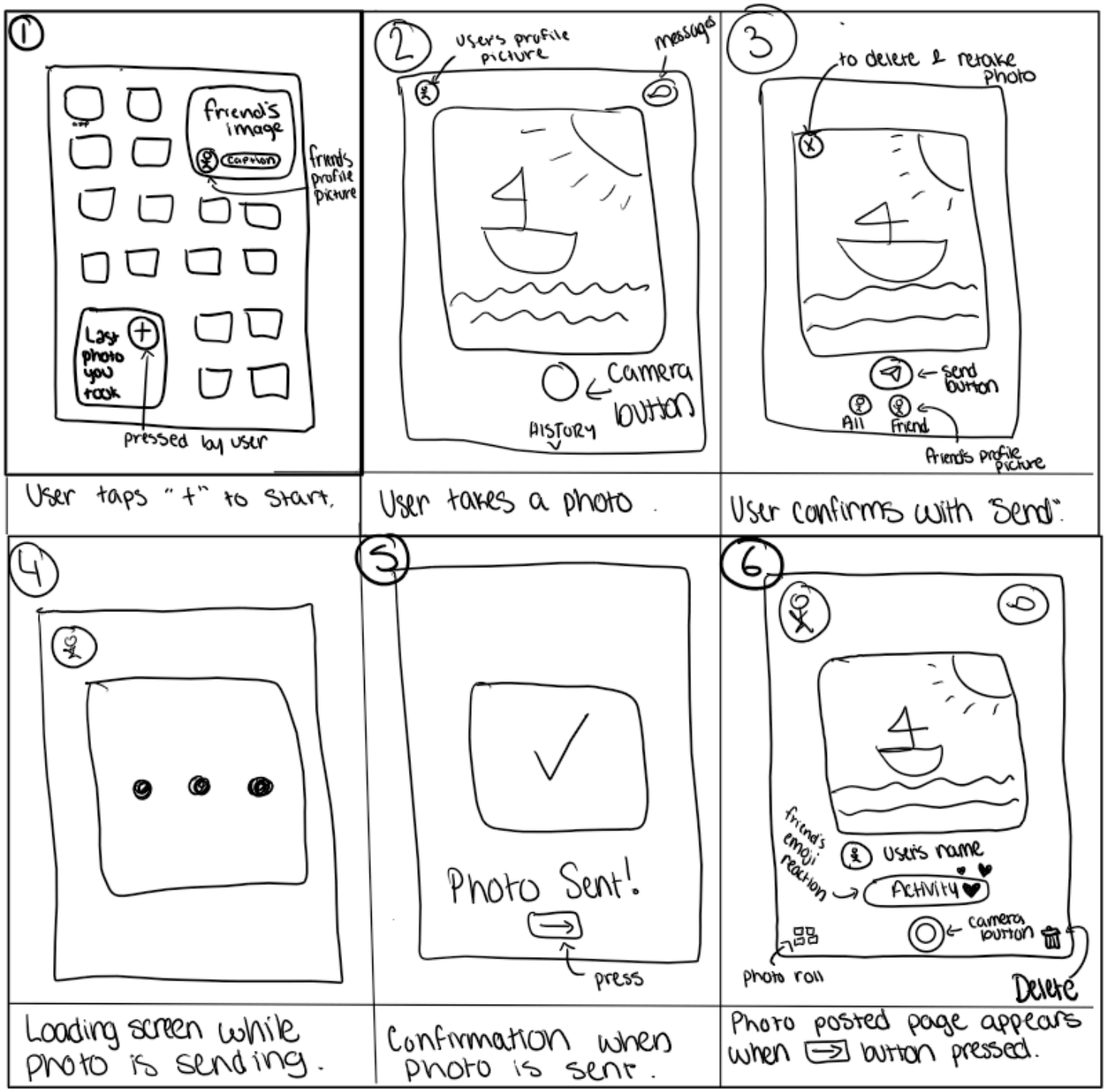
Starting assumptions for storyboard #1:

- The user already has Locket App installed and an account created.
- The user has at least one friend added.
- Start: The user starts on their homescreen on their mobile telephone.
- End: After the photo is sent, the goal is to share a photo quickly and confidently using the updated interface.

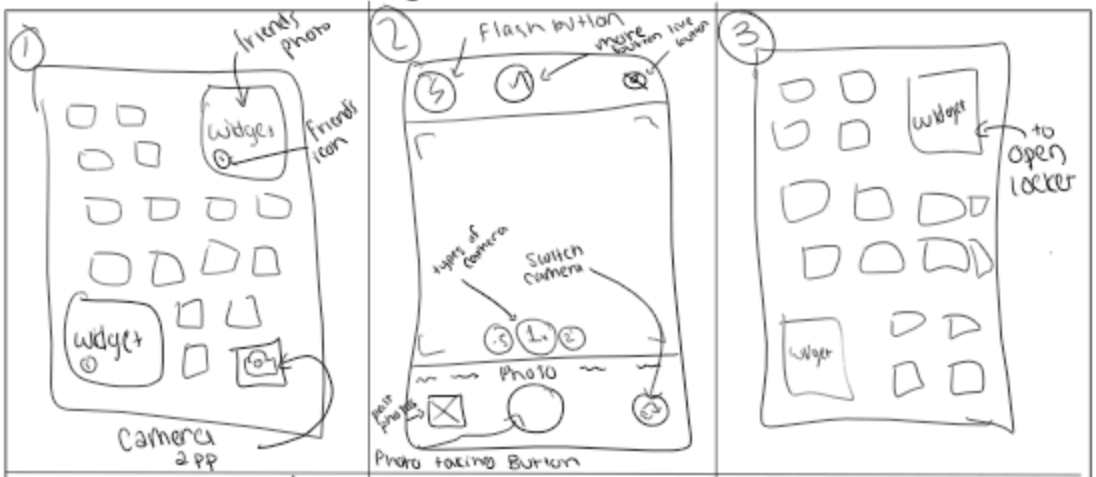
Starting assumptions for storyboard #2:

- The user already has Locket App installed and an account created.
- The user has at least one friend added.
- Start: The user starts on the homescreen of the app.
- End: After the photo is sent, the goal is to share a photo quickly and confidently using the updated interface.

Locket Storyboard #1



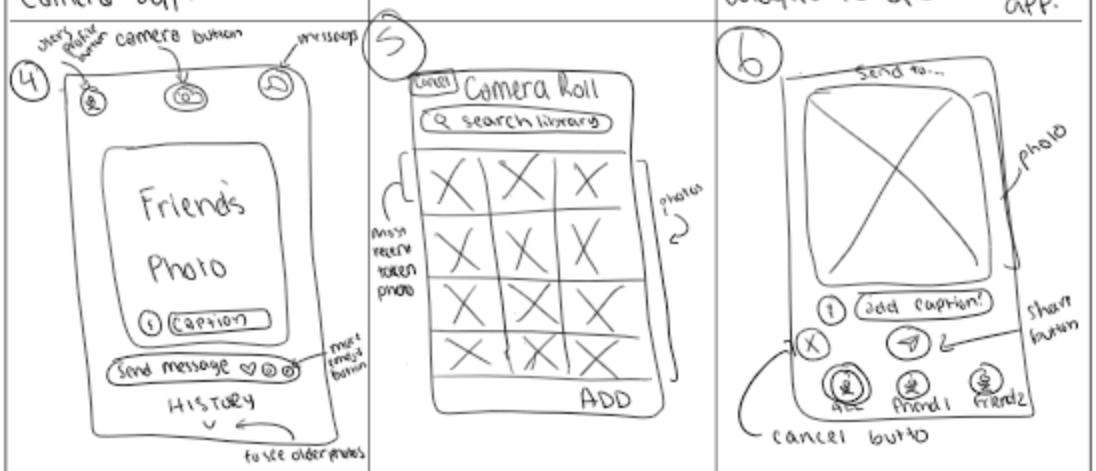
Locket Storyboard #2



User clicks pre-installed Camera app.

User presses "Photo Taking Button" to take photo.

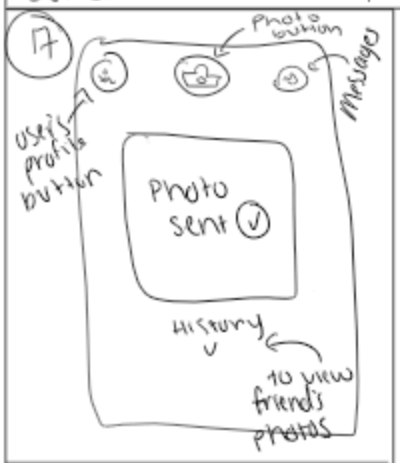
User clicks one of their widgets to open Locket app.



User clicks camera button.

User selects photo to share with friends.

User adds caption, picks who to send to, and shares photo.



Confirmation photo sent successfully.

ii. The storyboards have addressed all of the design problems that I have mentioned above, which includes, the lack of clear feedback, the difficulty of finding some features, the inconsistency with user expectations from other apps, and the overall reduction in user enjoyability and emotional connection. The original app's lack of clear feedback, which violated Shneiderman's principle and reduced effectiveness, is directly remedied in Storyboard #1 and Storyboard #2. This design explicitly incorporates a "Loading screen while photo is sending" and a "Confirmation when photo is sent." This offers continuous, informative feedback, clearly communicating the system's status at every stage of the interaction. By making the process visible, the design prevents user uncertainty and ensures users are never left wondering if their action, like sending a photo, was successful. This resolves the frustration seen in the user transcripts where unclear messages like "Not enough photos" caused hesitation.

Furthermore, to solve the problem of hidden features, for example, the delete option, I have added a trashcan icon to represent the delete button in Storyboard #1. This provides a strong visual affordance, making the function immediately obvious and accessible, unlike the original design which buried it under a "Share" menu. Specifically, the button is designed with a universal trash can icon. This uses the principle of perceived affordance, the icon is a visual cue that signals the button's function, to discard or remove a photo. Users instantly recognize the trash can symbol from both the physical world and countless other digital interfaces, allowing them to intuitively understand what the button does without needing to read the text or explore hidden menus. This direct mapping between the icon) and the intended action bridges the Gulf of Execution, making the interface feel more natural and significantly reducing the cognitive load and time previously spent searching for this essential feature.

To solve the capture error of accidentally activating premium features, the redesigned interface fundamentally changes the function of the contentious icon to align with user expectation. Previously, tapping what looked like a "photo gallery" icon would unexpectedly divert the user to a "Locket Gold" subscription page. This violated the consistency and standards principle, as it used a familiar symbol for a common function (reviewing photo roll) to perform an unrelated commercial action, thereby betraying the user's mental model built from countless other applications. The updated design in both storyboards ensures the icon now predictably opens the user's actual photo history, a chronological roll of photos they and their friends have shared. This directly adheres to perceived affordance, the icon now truthfully communicates its function, allowing users to correctly predict the outcome of their action. Furthermore, perceived affordances were used in both storyboards for all buttons, for example, the photo taking button, the cancel or delete button, the messages button, etc. By closing the gap between the user's intention (to reminisce or find an old photo) and the system's action (displaying the photo history), this change eliminates the frustrating capture error, reduces cognitive load, and restores a

sense of trust and control, which strengthens both learnability and the overall enjoyability of the app.

iii. I asked a friend to look at the two new Locket storyboards, and they had a lot of thoughts about the first one. Right away, they liked how it looked. They described the design as "clean" and "minimalistic" . They thought it was easy on the eyes and looked modern. They also found the steps in the app very clear and easy to follow. They specifically pointed out two small parts they really appreciated. First, they liked seeing a "Loading screen" because it would let them know the app is working and hasn't frozen. Second, they loved the "Photo Sent!" confirmation message. They said that these little details would make the app feel much more reliable and trustworthy, so you never have to wonder if your picture actually went through. Overall, their feeling was that Storyboard #1 wasn't just a small update, but a major and polished upgrade with many changes that fix the main problems they had previously encountered with the app. The main feedback that they had with this design is that the minimal style might be a bit plain for some users and feel like it lost its original friendliness of the app.

For Storyboard #2, they liked how flexible and familiar this version was. It's very flexible because it lets them use their phone's regular camera app. This means they can use all the features they are already familiar with, like the flash, different lenses, or filters, to take a really high-quality photo before sending it. The storyboard shows you can scroll through your "Photos" which is great because it means you aren't limited to just taking a new picture, you can quickly find and send a fun photo you already have saved on your phone or from another app. Since the user is just using the camera and photo gallery, they know exactly how everything works. There's no new, complicated camera inside the Locket app that you would have to learn. The main feedback they had about this storyboard is that it takes too many steps and is very overcomplicated. To send a single picture, you have to open your camera app and take the photo, then close the camera, find the Locket app, and open it to finally send the picture. This long process feels "slow and frustrating."

Out of the two storyboards, they preferred the first one since it felt simpler and fixed all the issues they previously had with the app. Based on this feedback, I have selected Storyboard #1 to move forward with. My friend's reaction confirms that the strength of this redesign lies in its focused improvements. By enhancing clarity and feedback through a minimalist aesthetic, it more effectively tackles the core problems of effectiveness and learnability without overcomplicating the user experience, which in turn creates a more solid foundation for long term enjoyability. Furthermore, with their feedback, when I get to the next phase of design, I will be able to add colours, and more personable things which will make the app feel more friendly and less generic.

b. Wireframe attached in separate document.

3.

a. Task: Sending a photo to a friend using the redesigned interface.

Step 1: Opening the Camera from the Homescreen	
1. Will the correct action be sufficiently evident to the user?	Yes - The large “+” icon is highly visible and located on the widget of a last photo taken from the user, indicating the plus means to take another photo. Note: While the icon has strong affordance, new users might be uncertain whether it opens the camera or sends a photo automatically. Adding a short tooltip or label such as “Tap to take photo” could reduce confusion. This aligns with Shneiderman’s principle of visibility
2. Will the user notice that the correct action is available?	Yes - The button has a large “+” icon which is salient and affordances are clear. Note: The button should maintain consistent contrast across light and dark modes to ensure accessibility. Reinforcing this with an animation or slight glow effect could further utilize salience.
3. Will the user associate the correct action with the task they are trying to achieve?	No - The “+” symbol could represent multiple functions, like adding friends or creating a new post. This violates Shneiderman’s principle of match between system and real world, since the mental model of taking a photo is more closely tied to a camera icon. Note: This indicates a mismatch in conceptual models. Adding a simple confirmation message such as “Send photo?” would improve error prevention and help users map their goals to the correct action.
4. Will the user interpret the response from the action correctly?	Yes - Tapping the icon opens the app to the camera view, providing immediate system feedback, confirming that the correct task was initiated. Note: N/A

Step 2: Capturing the Photo	
1. Will the correct	Yes - The circular button at the bottom center matches user

action be sufficiently evident to the user?	expectations from other camera apps (strong perceived affordance). Notes: N/A
2. Will the user notice that the correct action is available?	Yes - The capture button is large and central, visually separated from other icons, using Gestalt proximity effectively. Note: Good use of Gestalt's proximity and contrast principles ensures attention is drawn to the correct control.
3. Will the user associate the correct action with the task they are trying to achieve?	Yes - The mental model of "pressing the round button to take a photo" is universally familiar and connects to the mental model of taking a photo which minimizes cognitive effort. Note: This supports direct manipulation principles, allowing users to feel in control and confident about their actions.
4. Will the user interpret the response from the action correctly?	Yes - Once the button is pressed, the captured photo appears immediately, confirming the action's success. Note: Adding a shutter sound or quick flash animation could reinforce immediate feedback .

Step 3: Add caption and select recipient to send the photo	
1. Will the correct action be sufficiently evident to the user?	Yes - The caption field is clearly labeled, and the "Share" icon is a familiar visual metaphor. Note: This supports recognition over recall and provides clear affordance for sharing, but adding brief instructions ("Add caption and select who to send to") could strengthen learnability for new users.
2. Will the user notice that the correct action is available?	Yes - The "Share" button uses a universal logo and high contrast, making it easy to identify and it has an "Add Caption?" Note: The use of universal share icon aligns with Nielsen's principle of consistency and standards, helping users transfer knowledge from other apps.
3. Will the user associate the correct action with the task they are trying to achieve?	No - The "Send to All" option is located below the "Share" button, which could lead users to skip recipient selection and send unintentionally. Note: This violates error prevention principles. Reordering the

	interface so recipient selection appears before “Share” would reduce user errors and improve task flow consistency.
4. Will the user interpret the response from the action correctly?	Yes - The interface transitions to a loading screen followed by a confirmation message, signaling success. Note: The use of progressive disclosure provides ongoing feedback and supports closure, a key concept in Shneiderman’s design principles.

Step 4: Loading and confirmation screen	
1. Will the correct action be sufficiently evident to the user?	Yes - The “Photo Sent!” message is displayed prominently in the center, signaling task completion. Note: This aligns with Shneiderman’s principle of closure and reinforces the end of the task sequence, supporting user satisfaction.
2. Will the user notice that the correct action is available?	Yes - The message’s contrast and centered placement ensure visibility. Note: Use of Gestalt’s central focus principle ensures visibility, but adding a subtle animation could further emphasize confirmation.
3. Will the user associate the correct action with the task they are trying to achieve?	Yes - The confirmation directly references their action (“Photo Sent!”), providing clear mapping between intention and outcome. Note: This satisfies Nielsen’s match between system and real world, ensuring clarity and reducing ambiguity.
4. Will the user interpret the response from the action correctly?	Yes - The immediate confirmation and visual feedback (the heck mark) shows the system status, ensuring the user understands their photo was successfully sent. Note: This upholds Nielsen’s visibility of system status and fosters trust by confirming that the system executed the user’s request as intended.

b. Starting assumptions:

- The user's goal is to send one new photo.
- The Locket widget is visible on the phone's home screen.
- The hand starts on the phone and ends on the phone after the task is complete.

- The user will type a 5 letter caption
- Tapping on a button is modeled as a K operator (a keypress alternative).
- The user is average non-secretary typist (40 wpm)

Original Locket App	New Locket App
1. Initiate Task: Decide to send a new photo. - M -1.35s 2. Home Hand to Phone: Prepare to interact with the screen. - H - 0.40s 3. Find Locket Widget on the home screen. - M - 1.35s 4. Tap the Locket Widget. - K- 0.28s 5. Find the "Photo" Button inside the app. - M - 1.35 6. Tap the "Photo" button. - K - 0.28s 7. Find the "Caption" Text Field. - M - 1.35s 8. Tap the "Caption" field to focus it. - K- 0.28s 9. Making a strategy decision: Mental preparation to type the caption. - M - 1.35s 10. Type Caption: Enter 5 letters. - T(5) - 5 * 0.28 = 1.40s 11. Find the "Add Friends" Button. - M - 1.35s 12. Tap the "Add Friends" button. K- 0.28s 13. Find the "Share" or "Send" Button. - M - 1.35s 14. Tap the "Share" button. K- 0.28s 15. Verify from minimal feedback that the photo was sent. - M - 1.35s	1. IInitiate Task: Decide to send a new photo. - M -1.35s 2. Home Hand to Phone: Prepare to interact with the screen. - H - 0.40s 3. Find Locket Widget on the home screen. - M - 1.35s 4. Tap the Locket Widget. - K- 0.28s 5. Find the "Photo" Button inside the app. - M - 1.35 6. Tap the "Photo" button. - K - 0.28s 7. Find the "Caption" Text Field. - M - 1.35s 8. Tap the "Caption" field to focus it. - K- 0.28s 9. Making a strategy decision: Mental preparation to type the caption. - M - 1.35s 10. Type Caption: Enter 5 letters. - T(5) - 5 * 0.28 = 1.40s 11. Find the "Add Friends" Button. - M - 1.35s 12. Tap the "Add Friends" button. K- 0.28s 13. Find the "Share" or "Send" Button. - M - 1.35s 14. Tap the "Share" button. K- 0.28s 15. Verify from minimal feedback that the photo was sent. - M - 1.35s
Total Time: 14s	Total Time: 14s

Based on the KLM-GOMS analysis, the predicted time to finish the task is the same for the old design and the new one. However, this simple time comparison misses the main ways the new design is better. The most important improvement is that the new design stops users from making a common mistake. In the old app, tapping the photo icon could accidentally take you to a paid feature page, but the new design uses a clear "+" button that works correctly every time. The new layout also provides better confirmations, like a "Photo Sent!" message. This makes the app feel

more reliable and easier to use. So, the real benefit isn't that tasks are done much faster, but that they are done without confusion or errors, making the app more effective and enjoyable for the user.

4.

a. What can you say about sketching/storyboarding versus wireframing? Was it helpful to go through a storyboard first or would you have gone straight to wireframes?

Starting with sketching/storyboarding was a very helpful first step, and if I were to redo this assignment, I would still do this before the wireframing. The storyboard is like a simple comic strip that tells a story about a person using the app. It doesn't focus on small design details like exact button shapes or colors. Instead, it shows the user's situation, their goal, and the steps they take to achieve it. For example, my storyboard showed a user wanting to quickly share a fun moment with a friend. It illustrated them taking out their phone, finding the app, taking a photo. This process forced me to think about the user's feelings and the real world flow of the app. It helped me spot big problems early, like if a step was confusing or if the app didn't provide good feedback. By figuring this out first with simple drawings, I made sure the app's overall journey was logical and enjoyable before I spent time designing the actual screens.

After the storyboard was clear, then it was the right time to move on to wireframing. A wireframe is like a detailed blueprint or a skeleton for each screen. It takes the story from the comic strip and turns it into a practical layout. This is where I decided where to put the camera button, the box for typing a caption, and the send button. If I had jumped straight to making wireframes, I would have likely gotten stuck on small details, like making a button look perfect, without knowing if the overall process made sense. I might have designed a]screen that was part of a confusing journey. Since I did the storyboard first, I had a clear map to follow. When I created my wireframes, I could focus all my energy on making each screen easy to understand and use, confident that the path the user was taking was already a good one. So, both steps are important, but starting with the simple story makes the complex screen design much easier and more effective.

b. Which of the cognitive walkthrough or KLM-GOMS did you find to be more useful for initial evaluation?

I found the cognitive walkthrough to be more useful for the initial evaluation of my Locket app redesign. This method was more helpful because it forced me to see the app from the perspective of a new user who is still learning how it works, especially since one of my usability goals that I am focusing on is learnability. The walkthrough asks specific questions at each step and by answering these questions for my design, I could immediately spot confusing parts. For example,

it helped me realize that a button I thought was obvious might not be clear to someone using the app for the first time. This focus on learnability was perfect for finding and fixing problems early on.

In contrast, the KLM-GOMS model was less useful at this early stage. KLM-GOMS is great for measuring the speed of an expert user who already knows how to use the app perfectly. However, for my initial evaluation, my main concern wasn't timing an expert, it was making sure a beginner could figure out the app without getting lost or frustrated. The cognitive walkthrough directly addressed this goal by highlighting where users might get stuck, which allowed me to improve the design's clarity and flow before even thinking about its speed.